MIT805
Part 2: PySpark, MapReduce and Visualisation

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1 Introduction and Processing Objective

1.1 Dataset background

The New York City Yellow Taxi trip dataset contains individual taxi trip records with information about pickup and drop-off timestamps, pickup and drop-off location identifiers, passenger counts, trip distance, payment type, fare amount, tip amount, and total amount charged. The dataset was selected because its size and structure are suitable for demonstrating distributed data processing using PySpark.

The analysis is based on 24 months of Yellow Taxi trip data for the 2016 to 2017 period. The original dataset occupied approximately **3.2** GB. The data was processed using Apache Spark rather than converting the complete dataset to Pandas. Pandas was used only after the PySpark processing had reduced the results to a small, aggregated table required for visualisation.

1.2 Problem statement

Taxi demand is not distributed evenly across New York City. Some locations experience frequent passenger demand, while other locations may produce fewer trips but generate greater revenue because of longer or more valuable journeys. Demand may also change according to the hour of the day.

Understanding these differences can help taxi operators allocate vehicles more effectively, assist drivers in identifying demand patterns, and support transport authorities in planning services around high activity locations.

1.3 Central analytical question

The central analytical question is:

> **How do Yellow Taxi demand and revenue differ across New York City pickup zones and time periods, and which locations and periods contribute most to taxi activity and earnings?**

1.4 Processing objectives

The objectives of the analysis were to:

- clean invalid and incomplete taxi trip records using PySpark
- transform timestamps into useful date, and day variables

- measure taxi demand using the number of valid trips
- calculate total revenue by pickup zone
- calculate average trip distance and duration by pickup zone
- identify the busiest and highest revenue pickup zones
- analyse changes in taxi demand across the times of the day
- demonstrate mapping, grouping, shuffling, and reduction operations
- provide evidence of Spark's distributed execution model
- interpret the findings from business, and societal perspectives.

2 Data ingestion and cleaning

During the ingestion and cleaning phase, monthly NYC Yellow Taxi Parquet files from 2016 and 2017 were loaded into Spark DataFrames using `spark.read.parquet()`, and only the columns required for the analysis were selected to improve efficiency. A number of new features were derived, including `trip_duration_minutes`, `day_of_week`, `month`, `pickup_hour`, and `time_of_day`. Invalid values where `trip_distance`, `fare_amount`, or `passenger_count` were less than or equal to zero are removed. The processed monthly datasets were combined using `unionByName()` and then joined with the taxi zone lookup table using `PULocationID` to add meaningful zone and borough information. After cleaning, **242824688** valid records remained.

3 Mapping and Transformation

The mapping stage selected the attributes required for the analysis and transformed them into an analysis-ready dataset. Zone and borough information were renamed to `pickup_zone` and `pickup_borough`, while attributes such as distance, duration, fare, tip, total amount, and passenger count were retained.

Listing 1: Mapping taxi trip records

```
mapped_df = (
    df_all
    .select(col("Zone").alias("pickup_zone"),
            col("Borough").alias("pickup_borough"),
            "time_of_day",
            "day_of_week",
            "trip_distance",
            "trip_duration_minutes",
            "fare_amount",
            "tip_amount",
            "total_amount",
            "passenger_count"
    )
)
```

The `select()` operation is a lazy transformation. Spark records the transformation in its execution plan but does not execute it until an action is triggered. The resulting dataset contained the key attributes required for grouping and aggregation operations.

4 Shuffle and Grouping

The mapped records were grouped using the composite key consisting of `pickup_zone`, `pickup_borough`, and `time_of_day`.

Listing 2: Composite grouping records by multiple keys

```
summary_agg_df = (
    mapped_df
    .groupBy(
        "pickup_zone",
        "pickup_borough",
        "time_of_day"
    )
    .agg(
        count("*").alias("trip_count"),
        round(sum("total_amount"),2).alias("total_revenue"),
        round(avg("trip_distance"),2).alias("average_trip_distance"),
        round(avg("trip_duration_minutes"),2).alias("
            average_trip_duration")
    )
    .orderBy("total_revenue", ascending=False)
)
```

This aggregation was performed using Spark's `groupBy()` operation to calculate demand, revenue, and trip characteristics for each location and time-of-day combination.

The grouping operation introduced a wide dependency because records belonging to the same group could be distributed across multiple partitions. Spark therefore redistributed records across executors so that all records sharing the same composite key were brought together for aggregation.

This redistribution phase is known as a shuffle operation. The shuffle was necessary because each executor could initially compute only partial aggregates from its local partitions. To produce the final results, Spark exchanged data between executors and merged the partial aggregates. Shuffling is one of the most expensive operations in Spark because it may involve network communication, and key-based sorting.

5 Reduction and Aggregation

After the shuffle stage, Spark performed the reduction phase by applying aggregate functions to each group. The `count()` function calculated the number of trips (`trip_count`). The `sum()` function was used to calculate total revenue (`total_revenue`), while the `avg()` function calculated average trip distance (`average_trip_distance`) and average trip duration (`average_trip_duration`). These aggregated metrics provided insight into the busiest pickup locations, the revenue generated in different areas, and how trip characteristics varied across times of the day. After aggregation, **784** valid records remained, which is extremely smaller compared to the original cleaned dataset.

6 Spark Execution and DAG

Spark execution evidence was obtained using:

```
summary_agg_df.explain()
```

Unlike Hadoop MapReduce, which follows a fixed

```
Map -> Shuffle -> Reduce
```

process, Spark builds a Directed Acyclic Graph (DAG) of transformations. Spark optimises the execution plan before creating stages and tasks.



Figure 1: Spark DAG Execution

7 Results

Table 1 shows the top pickup zones, boroughs and time of day, with the highest demand and revenue.

Table 1: Top 20 Locations with the highest Trip Demand and Revenue

Pickup Zone	Borough	Time of Day	Trip Count	Revenue ($)	Avg Distance (miles)	Avg Duration (mins)
JFK Airport	Queens	Night	2,534,493	140,539,019	19.36	37.65
JFK Airport	Queens	Day	1,910,263	110,820,532	20.89	36.44
LaGuardia Airport	Queens	Night	2,585,807	104,446,945	9.64	28.47
LaGuardia Airport	Queens	Day	2,222,646	99,148,894	12.23	37.50
East Village	Manhattan	Night	4,816,597	71,527,653	2.86	16.05
LaGuardia Airport	Queens	Morning	1,448,667	66,938,133	10.33	41.41
Times Sq/Theatre District	Manhattan	Night	4,179,367	62,720,399	2.69	16.20
JFK Airport	Queens	Morning	1,130,966	62,710,783	18.24	49.38
Clinton East	Manhattan	Night	3,824,318	57,130,235	4.12	15.57
Midtown Center	Manhattan	Night	3,870,792	55,871,025	2.38	15.17
Union Sq	Manhattan	Night	3,889,973	55,294,580	2.35	15.04
Midtown East	Manhattan	Night	3,540,381	51,808,092	2.46	14.76
Penn Station/Madison Sq West	Manhattan	Night	3,326,228	50,877,618	9.47	16.73
Midtown Center	Manhattan	Day	3,196,587	49,870,562	2.53	17.95
Murray Hill	Manhattan	Night	3,180,112	46,147,635	2.44	14.80
West Village	Manhattan	Night	3,067,163	45,340,371	5.36	15.83
Lower East Side	Manhattan	Night	2,805,629	44,801,091	3.05	17.09
Upper East Side South	Manhattan	Day	3,674,795	44,518,845	1.70	13.77
East Chelsea	Manhattan	Night	2,934,626	43,366,357	5.51	15.63
Midtown North	Manhattan	Night	2,937,085	42,585,651	2.66	15.51

The results show that taxi demand and revenue were highly concentrated in a small number of locations. Airport zones, particularly JFK Airport and LaGuardia Airport, generated the highest revenue despite having longer average trip distances and durations. The results also show the influence of time of day on taxi activity. Most of the highest performing locations recorded their strongest demand and revenue during the night period.

8 Visualisation

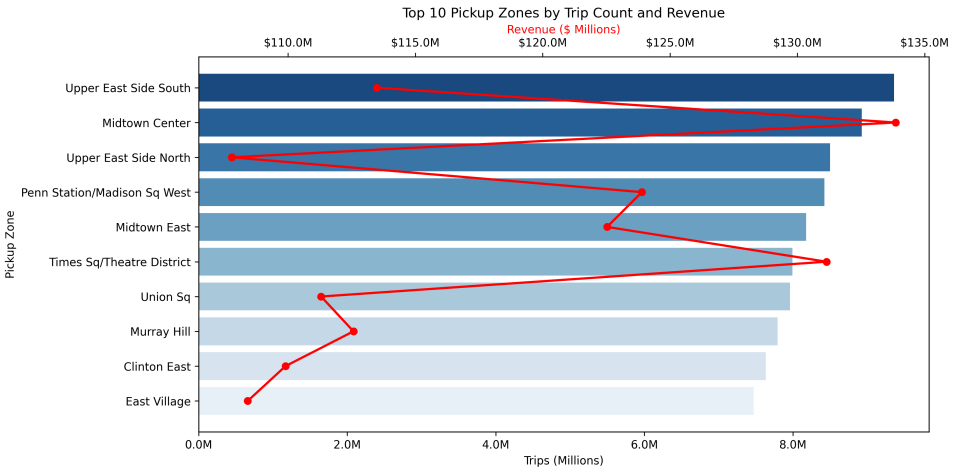


Figure 2: Top 10 Pickup Zones by Number of trips and Revenue

Figure 2 shows the top 10 pickup zones by trip demand and total revenue. It can be observed that the zone with the highest demand does not generate the highest revenue. While Midtown Center records the second largest number of trips and the highest revenue, other high demand zones generate comparatively less revenue.This suggests that trip volume alone does not determine revenue generation.

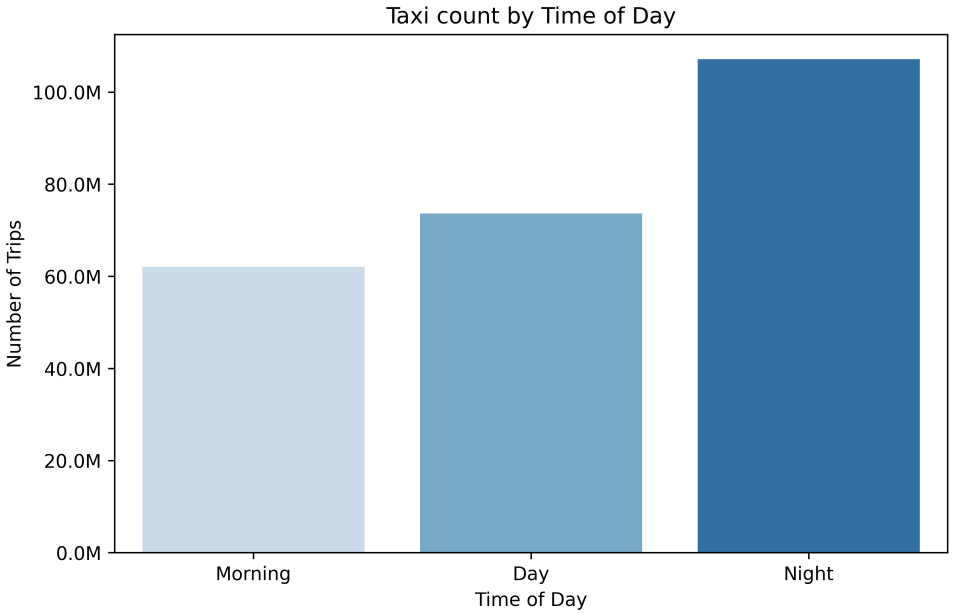


Figure 3: Number of trips by Time of day

Figure 3 shows the distribution of taxi trips by time of day. It can be observed that most trips occur during the night period. This suggests that taxi services are used more frequently during night hours, possibly because of nightlife activities, safety concerns or reduced availability of public transport such as buses and subways. In contrast, fewer trips were recorded during the morning and daytime periods.

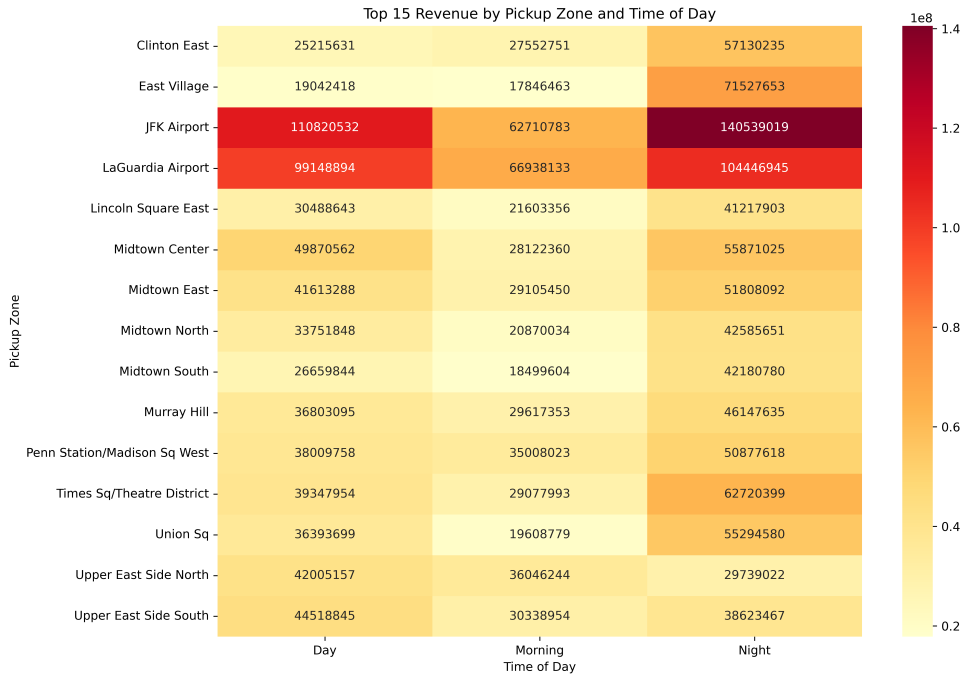


Figure 4: Top 15 Revenue by Pickup Zone and Time of day

Figure 4 shows the top 15 revenue generating pickup zones across different times of the day. The heatmap indicates that JFK Airport generated the highest revenue during both the night and day periods, which highlights its importance as a major source of taxi revenue. LaGuardia Airport also generated substantial revenue across all time periods, particularly during the night.

9 Business and Societal Interpretation

9.1 Value to taxi operators and drivers

The concentration of demand in the Upper East Side South zone suggests that taxi operators could position additional vehicles near these areas before the identified peak periods. For example, the high demand observed at night could support an increase in vehicle availability shortly before that period.

The comparison between high volume zones and high revenue zones is also important. Drivers seeking frequent passenger trips may prioritise zones with high trip counts. Drivers or operators focusing on trip value may consider zones with higher average revenue and longer average distances. These findings could reduce unproductive vehicle movement, but oversupplying the same locations could create additional congestion and reduce the expected benefit.

9.2 Value to passengers and transport authorities

Better alignment between vehicle supply and passenger demand could reduce waiting times in high-demand locations. Authorities could also use the identified hotspots to review taxi pickup areas, and coordination with tourism, and airport services.

The analysis may contribute to more informed urban transport planning by showing where taxi activity is concentrated. Reduced empty vehicle movement could potentially reduce unnecessary fuel consumption. However, the dataset does not directly measure emissions, congestion,

or distance travelled without passengers. The environmental value should therefore be treated as a possible implication rather than a confirmed causal outcome.

9.3 Limitations

The analysis has several limitations:

1. **Historical coverage:** The results describe the selected historical period and may not represent future demand.
2. **External variables:** Weather, public events, road closures, traffic conditions, and public transport disruptions were not included.
3. **Transport coverage:** The dataset covers Yellow Taxi trips and does not represent all ride hailing, private vehicle, rail, bus, or pedestrian activity.
4. **Data quality:** Cleaning rules may have excluded some genuine unusual trips.
5. **Revenue interpretation:** The total amount represents recorded trip revenue rather than business profit. Operating costs were not available.

10 Conclusion

This project demonstrated distributed processing using PySpark and the MapReduce paradigm. Data was mapped, shuffled, grouped and reduced to identify the pickup locations that generated the highest levels of taxi demand and revenue. The results showed that demand is concentrated in specific zones and varies across time periods. These insights can support operational and transport planning decisions.

Project Links

GitHub repository: https://github.com/Siya1314/MIT-805-Project

References

[1] New York City Taxi and Limousine Commission. *TLC Trip Record Data*. Available at: https://www.nyc.gov/site/tlc/about/tlc-trip-record-data.page.